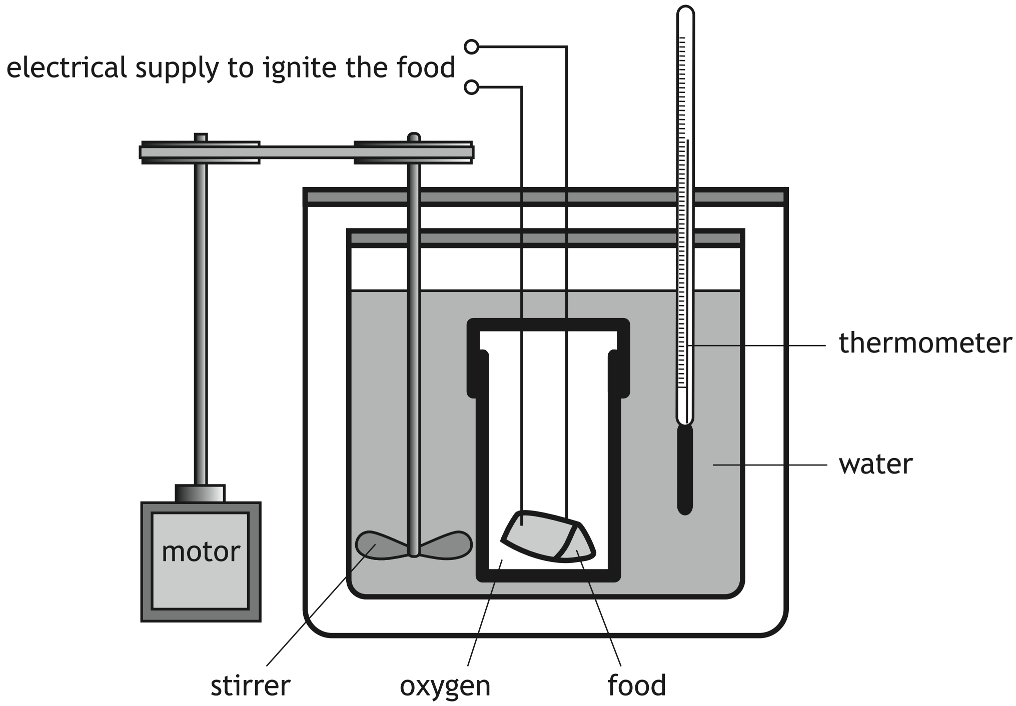
MARKS | DO NOT WRITE IN THIS MARGIN

5. A bomb calorimeter is used in the food industry to determine the energy released by foods when they are burned.



Bomb calorimeter

(a) Identify a feature of the bomb calorimeter and explain how this allows the accurate determination of the energy released by burning foods. **1**

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5. (continued)

(b) Fats and oils release a large amount of energy when they are burned.

(i) A 1·00 g sample of the oil, triolein ($GFM = 884$ g) was burned in a bomb calorimeter.

The temperature rise in the 775 cm^3 of water was 11·9 °C.

Calculate the enthalpy of combustion, in kJ mol^{-1}, of triolein. **3**

(ii) Foods with a lower respiratory quotient are better for people who find it difficult to obtain energy from food.

The respiratory quotient, RQ, is the ratio of carbon dioxide, CO_2, produced to the oxygen, O_2, consumed when a food is burned in the body.

$$\text{Respiratory quotient} = \frac{CO_2 \text{ produced}}{O_2 \text{ consumed}}$$

The equation for the combustion of triolein, $C_{57}H_{104}O_6$, is shown.

$$C_{57}H_{104}O_6(\ell) \quad + \quad 80\ O_2(g) \quad \rightarrow \quad 57\ CO_2(g) \quad + \quad 52\ H_2O(\ell)$$

Determine the respiratory quotient for triolein. **1**

5. (continued)

(c) Tristearin, $C_{57}H_{110}O_6$, is a saturated fat.

The table shows the viscosity of different saturated fats at 70 °C.

Fat	Molecular formula	Viscosity at 70 °C (units)
Tributyrin	$C_{15}H_{26}O_6$	3·0
Tricaproin	$C_{21}H_{38}O_6$	5·9
Tricaprylin	$C_{27}H_{50}O_6$	8·8
Tricaprin	$C_{33}H_{62}O_6$	11·7
Trilaurin	$C_{39}H_{74}O_6$	14·6

(i) Predict the viscosity of tristearin at 70 °C. **1**

(ii) Edible fats and oils are molecules that contain three ester links.

Explain why glycerol is able to form fats and oils. **1**

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